AMAZON PRIME | PROJECT REPORT DOCUMENT

Amazon Prime Cart Abandonment & Customer Segmentation Analysis

Analysing Cart Abandonment, Customer Segments and Revenue at Risk Using Excel and SQL

Internship Project Report

Tools Used: Microsoft Excel, PostgreSQL and VS Code

Internship Project Report	
Submitted by:	Mahi Choudhary
College/University	Rabindranath Tagore University, Bhopal
Internship Organization:	Internmo Edtech Pvt.Ltd.
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1. Project Overview

Amazon receive millions of visitors, but not every visitor completes a purchase. Some customers browse products, add items to their cart, or reach the checkout page and leave without buying anything.

The purpose of this project is to analyse shopping behaviour and understand where customers abandon their shopping sessions on amazon. Using Microsoft Excel and PostgreSQL, the dataset was cleaned and analysed to identify shopping patterns, customer segments, and potential revenue at risk.

The analysis also includes an interactive Excel dashboard to present important findings and recommendations to help improve the checkout experience and encourage more customers to complete their purchases.

2. Project Objectives

The main objectives of this project are:

1. To analyse amazon shopping session data and understand customer shopping behaviour.
2. To identify the stages where customers leave without completing their purchases.
3. To calculate the cart abandonment rate and identify potential revenue at risk.
4. To divide customers into different segments based on their shopping activity.
5. To use Excel and SQL to clean, organise and analyse the dataset.
6. To create an interactive Excel dashboard to present important findings.
7. To suggest practical ways to improve the checkout experience and encourage customers to complete their purchases.

3. Dataset Description

For this project, a simulated Amazon shopping behaviour dataset was provided for analysis. The dataset contained information about customer shopping sessions, including customer details, device types, traffic sources, session duration, cart values and the stage at which customers left the shopping process.

The original dataset contained 1,000 records. After removing **15 duplicate records**, 985 records remained for analysis.

The dataset included the following important fields:

- Session ID and User ID
- Session Date
- Prime Member Status
- Device Type
- Traffic Source
- Session Duration
- Items in Cart
- Total Cart Value
- Drop-off Stage
- Customer Segment

This dataset was used to understand customer behaviour, identify cart abandonment patterns, calculate important business KPIs and estimate potential revenue at risk.

Session_ID	User_ID	Session_Date	Prime_Member_Status	Device_Type	Traffic_Source	Session_Duration_Mins	Items_in_Cart	Total_Cart_Value	Drop_off_Stage	Data_Quality_Flag	Outlier_Analysis	Customer_Segg
S586221	U222579	10-10-2025	Prime	Mobile App	Unknown	5.9	1	40.17	Cart	Valid	Normal	Low-Value Aba
S837149	U339444	31-10-2025	Non-Prime	Desktop	Paid Search	17.4	1	23.55	Success	Valid	Normal	Purchasers
S886563	U192272	02-10-2025	Prime	Mobile App	Paid Search	4.8	0	0.00	Product Page	Valid	Normal	Browsers
S535129	U302760	18-10-2025	Non-Prime	Mobile App	Direct	2.3	0	0.00	Home Page	Valid	Normal	Browsers
S677101	U991879	19-10-2025	Non-Prime	Mobile App	Organic Search	15.9	2	82.12	Success	Valid	Normal	Purchasers
S963111	U663401	25-10-2025	Non-Prime	Mobile Web	Affiliate	10.6	3	75.55	Cart	Valid	Normal	Mid-Value Abar
S578343	U515168	09-10-2025	Prime	Desktop	Paid Search	9.1	1	73.71	Cart	Valid	Normal	Mid-Value Abar
S579308	U972140	11-10-2025	Prime	Mobile App	Organic Search	19.0	1	8.55	Success	Valid	Normal	Purchasers
S862553	U798774	26-10-2025	Prime	Tablet	Organic Search	8.3	3	56.88	Cart	Valid	Normal	Mid-Value Abar
S883245	U319269	28-10-2025	Non-Prime	Tablet	Organic Search	2.7	0	0.00	Product Page	Valid	Normal	Browsers
S697066	U353602	17-10-2025	Prime	Tablet	Affiliate	1.7	0	0.00	Product Page	Valid	Normal	Browsers
S649807	U952211	10-10-2025	Non-Prime	Mobile App	Paid Search	8.6	1	46.73	Cart	Valid	Normal	Low-Value Aba
S716722	U593418	17-10-2025	Prime	Mobile App	Affiliate	8.5	2	40.53	Payment	Valid	Normal	Other
S616555	U233576	18-10-2025	Non-Prime	Mobile App	Direct	17.0	3	53.03	Success	Valid	Normal	Purchasers
S697262	U688016	20-10-2025	Non-Prime	Desktop	Email	25.6	3	56.31	Success	Valid	Normal	Purchasers
S635973	U178772	13-10-2025	Non-Prime	Mobile App	Direct	8.3	1	16.12	Cart	Valid	Normal	Low-Value Aba
S822617	U959152	25-10-2025	Non-Prime	Mobile Web	Direct	7.2	2	186.08	Cart	Valid	Potential Outlier	High-Value Aba
S748517	U428345	08-10-2025	Prime	Desktop	Paid Search	19.7	2	52.91	Success	Valid	Normal	Purchasers
S672609	U140305	19-10-2025	Prime	Mobile App	Organic Search	17.8	2	38.88	Checkout	Valid	Normal	Other
S964548	U459907	29-10-2025	Non-Prime	Mobile Web	Organic Search	3.6	2	60.02	Cart	Valid	Normal	Mid-Value Abar
S715746	U197169	13-10-2025	Prime	Mobile Web	Email	10.4	1		Success	Valid	Normal	Purchasers
S983848	U396238	08-10-2025	Prime	Mobile App	Unknown	16.7	3	135.23	Success	Valid	Normal	Purchasers
S617408	U839289	06-10-2025	Prime	Mobile App	Direct	10.5	3	42.69	Cart	Valid	Normal	Low-Value Aba
S907212	U822750	29-10-2025	Non-Prime	Mobile App	Direct	6.3	1	14.51	Cart	Valid	Normal	Low-Value Aba
S614837	U426650	29-10-2025	Non-Prime	Mobile App	Direct	7.1	3	65.87	Cart	Valid	Normal	Mid-Value Abar
S992566	U937368	26-10-2025	Prime	Mobile App	Email	4.1	1	19.82	Cart	Valid	Normal	Low-Value Aba
S626404	U682961	24-10-2025	Prime	Desktop	Direct	14.1	4	101.91	Success	Valid	Normal	Purchasers
S598556	U559254	15-10-2025	Prime	Desktop	Direct	3.2	1	38.23	Cart	Valid	Normal	Low-Value Aba

Figure 1: Sample of the cleaned shopping behaviour dataset.

4. Data Cleaning and Preparation

Before starting the analysis, the dataset was checked and cleaned to improve its quality and consistency.

The original dataset contained 1,000 records. During the cleaning process, 15 duplicate records were identified and removed, leaving 985 records for further analysis.

Sr. No	Column Name	Issue Found	Action Taken	Records Affecte
1	Prime_Member_Status	Missing Values	Replaced blanks With Unknown	11
2	Device_Type	Missing Values	Replaced blanks With Unknown	20
3	Traffic_Source	Missing Values	Replaced blanks With Unknown	41
4	Session_Duration_Mins	Missing Values	Left blank for analysis	31
5	Total_Cart_Value	Missing Values	Left blank for analysis	18
6	Duplicate Records	15 Duplicate records found	Removed duplicate records	15
7	Total_Cart_Value	Negative Values found	Flagged for investigation,original records retained	3
8	Data_Quality_Flag	Negative Total_Cart_Value values	Added a flag to identify negative cart values; original values retained	3
9	Outlier_Flag	Potential outliers in Total_Cart_Value	Created a separate worksheet to calculate outlier limits using the IQR method	0
10	Total_Cart_Value	36 potential outliers identified using IQR method	Flagged for investigation; original values retained	36
11	Total_Cart_Value	Zero items with positive cart value	No issue found	0
12	Traffic_source	Two different names for one source.	Replaced case sensitive words with original ones	16
13	Drop_off_Stage	Two different names for one stage in different cases.	Replaced case sensitive words with original ones	107
14	Device_Type	Two different names for one stage in different cases.	Replaced case sensitive words with original ones	23
15	Session_date	Not in correct date format	Changed date format to English(India)date format.	985
16	Session_Duration_Mins	Unusually long session durations	Flagged for investigation; original values retained	6

Figure 2: Cleaning_Log sheet showing the cleaning activities.

The following cleaning activities were performed:

1. **Duplicate Removal:** Duplicate records were removed to avoid counting the same shopping session more than once.
2. **Missing Value Handling:** Missing values were identified in columns such as Prime Member Status, Device Type, Traffic Source, Session Duration and Total Cart Value. Missing categorical values were replaced with "Unknown", while missing numerical values were retained as blank rather than replaced with zero.

Column Names	Missing Values	Missing %
Prime_Member_Status	11	1.1%
Device_Type	20	2.0%
Traffic_Source	41	4.1%
Session_Duration_Mins	31	3.1%
Total_Cart_Value	18	1.8%
Total	121	2.42%

Figure 3: Missing_Values_Report sheet.

3. **Data Formatting:** Date and numerical columns were formatted appropriately in Excel to make the data easier to read and analyse.

4. **Consistency in Text Values:** In the Traffic Source, Device Type and Drop-off Stage columns, some values appeared in different letter cases, such as "direct" and "DIRECT". These inconsistent entries were standardised so that the same category was represented uniformly, avoiding duplicate categories and improving the accuracy of the analysis.

5. **Consistency Checks:** Categorical values such as Device Type, Prime Member Status, Traffic Source and Drop-off Stage were checked for consistency.

6. **Outlier Analysis:** The Interquartile Range (IQR) method was used to identify unusually high cart values. A total of 36 potential outliers were flagged for review.

Metric	Value
Q1	0
Q3	74.005
IQR	74.005
LOWER BOUND	-111.008
UPPER BOUND	185.0125

Figure 5: Outlier_Analysis sheet showing the IQR method and flagged records.



Figure 4: Standardisation of inconsistent text values in Traffic Source, Device Type and Drop-off Stage columns.

7. **Unusually Long Session Durations:** During the data quality check, six shopping sessions were found to have unusually long durations compared with typical sessions. These records were flagged for further investigation. The original values were retained because unusually long sessions should not be removed or modified without verification.

8. **Negative Value Checks:** Three records containing negative cart values were identified and flagged for investigation. The original values were retained rather than changed without verification.

9. **Customer Segmentation:** Customers were grouped into categories such as Browsers, Purchasers, Low-Value Abandoners, Mid-Value Abandoners and High-Value Abandoners based on their shopping activity.

After these steps, the cleaned dataset was used for further analysis in Excel and PostgreSQL.

5. Data Analysis Using Excel and SQL

After completing the data cleaning process, the dataset was analysed using Microsoft Excel and PostgreSQL to understand customer shopping behaviour and identify cart abandonment patterns.

Analysis Using Microsoft Excel

Excel was used to:

- Calculate important business KPIs, including conversion rate and cart abandonment rate.
- Analyse shopping funnel drop-offs at different stages.
- Calculate average abandoned cart value and potential revenue at risk.
- Group customers into different segments based on their shopping activity.
- Compare shopping behaviour between Prime and Non-Prime customers.
- Create an interactive dashboard to present the results.

Analysis Using PostgreSQL

PostgreSQL was used to:

- Import and organise the cleaned dataset.
- Write SQL queries to calculate business KPIs.
- Analyse cart abandonment and shopping funnel patterns.
- Compare customer behaviour across different groups.
- Verify the results obtained in Excel.

  The results from Excel and PostgreSQL were compared to check the accuracy and consistency of the analysis.

Funnel Drop-off Rate(Home Page)	10.05%	Percentage of sessions that ended at the Home Page without proceeding to the Product Page.
Funnel Drop-off Rate(Product Page)	24.60%	Percentage of sessions that ended at the Product Page without proceeding to the Cart.
Funnel Drop-off Rate(Cart)	26.65%	Percentage of sessions that ended at the Cart without proceeding to the Checkout.
Funnel Drop-off Rate(Checkout)	20.41%	Percentage of sessions that ended at the Checkout without proceeding to the Payment.
Funnel Drop-off Rate(Payment)	14.87%	Percentage of sessions that ended at the Payment without proceeding to the Success.

Amazon_data_analysis: WITH funnel AS (...

stage	drop_off_sessions	sessions_reache...	drop_off_rate
Home Page	99	985	10.05
Product Page	218	886	24.60
Cart	178	668	26.65
Checkout	100	490	20.41
Payment	58	390	14.87

Figure 6: KPI calculations using Microsoft Excel. | Figure 7: Shopping behaviour analysis using PostgreSQL.

6. KPI Results and Analysis

The cleaned Amazon shopping behaviour dataset was analysed to calculate key performance indicators (KPIs). These metrics helped identify customer conversion patterns, cart abandonment and potential revenue at risk.

The major findings are presented below:

Key Performance Indicator	Result
Total Shopping Sessions	985
Completed Purchases	332
Overall Conversion Rate	33.71%
Cart Abandonment Rate	26.65%
Checkout and Payment Abandonment Rate	32.24%
Average Abandoned Cart Value	$208.60
Potential Revenue at Risk	35,879.60
Average Successful Session Duration	22.55 minutes
Average Abandoned Session Duration	9.89 minutes

The analysis showed that 33.71% of shopping sessions resulted in completed purchases. The cart abandonment rate was 26.65%, indicating that some customers left after adding products to their carts.

The total cart value associated with abandoned cart sessions was **$35,879.60**. This represents potential revenue at risk, not guaranteed recoverable revenue.

Successful sessions lasted longer on average than abandoned sessions, suggesting a difference in browsing and purchasing behaviour.

KPI Name	Result	Explanation
Total Sessions	985	Total sessions including abandoned and successful.
Overall Conversion Rate	33.71%	Percentage of total sessions that resulted in a successful purchase.
Cart Abandonment Rate	26.65%	Percentage of sessions that reached the Cart stage but ended without proceeding further.
Funnel Drop-off Rate(Home Page)	10.05%	Percentage of sessions that ended at the Home Page without proceeding to the Product Page.
Funnel Drop-off Rate(Product Page)	24.60%	Percentage of sessions that ended at the Product Page without proceeding to the Cart.
Funnel Drop-off Rate(Cart)	26.65%	Percentage of sessions that ended at the Cart without proceeding to the Checkout.
Funnel Drop-off Rate(Checkout)	20.41%	Percentage of sessions that ended at the Checkout without proceeding to the Payment.
Funnel Drop-off Rate(Payment)	14.87%	Percentage of sessions that ended at the Payment without proceeding to the Success.
Checkout and Payment Abandonment	32.24%	Percentage of sessions that dropped off at the Product Page, according to the calculation previously performed.
Average Abandoned Cart Value	$208.60	Average value of shopping carts abandoned by customers at the Cart stage.
Revenue at Risk	$35,879.60	Total value of abandoned carts where customers left at the Cart stage without proceeding to checkout. It represents potential revenue that could be recovered by improving the shopping experience.
Prime Vs Non-Prime Gap	6.45%	Difference between the abandonment rates of Non-Prime and Prime customers.
Average Successful sessions duration by outcome	22.55	Successful purchase sessions lasted an average of 22.55 minutes.
Average Abandoned session duration by outcome	9.89	Abandoned sessions lasted an average of 9.89 minutes.
Average Difference (Successful minus Abandoned) session duration by outcome	12.66	Successful purchase sessions lasted an average of 12.66 minutes longer than abandoned sessions.

Figure 8 : Key performance indicators calculated using Excel.

7.Customer Segmentation Analysis

Customer segmentation was performed to understand different types of shoppers and their shopping behaviour. Based on the number of items in the cart, cart value and final shopping stage, customers were divided into five main segments.

The segments used in this project were:

1. **Browsers:** Customers who visited the website but did not add any items to their cart.
2. **Purchasers:** Customers who successfully completed their purchases.
3. **Low-Value Abandoners:** Customers who left their carts with a cart value below 50.
4. **Mid-Value Abandoners:** Customers who abandoned their carts with a value between 50 and 149.99.
5. **High-Value Abandoners:** Customers who abandoned their carts with a value of 150 or more.

This segmentation helped identify different customer groups and understand which groups may need more attention to improve purchase completion.

Customer segment	Number of sessions
Purchasers	332
Other	158
Browsers	317
Low-value Abandoners	85
Mid-value Abandoners	78
High-value Abandoners	15
Total	**985**

Figure 9: Customer segmentation based on shopping behaviour.

8. Shopping Behaviour Dashboard

An interactive dashboard was created in Microsoft Excel to present the key findings of the Amazon shopping behaviour analysis in a simple and easy-to-understand format.

The dashboard includes four key performance indicators: Total Sessions, Conversion Rate, Cart Abandonment Rate and Revenue at Risk.

It also contains three charts:

- **Shopping Conversion Funnel:** Shows how shopping sessions progress through different stages, from the Home Page to successful purchases.
- **Customer Segment Breakdown:** Displays the distribution of customers across different shopping segments.
- **Prime vs Non-Prime Shopping Behaviour:** Compares shopping behaviour across Prime and Non-Prime customers.

Interactive slicers for Device Type and Traffic Source allow users to explore the data from different perspectives.

The dashboard helps present important business insights and makes it easier to identify areas where the shopping experience can be improved.

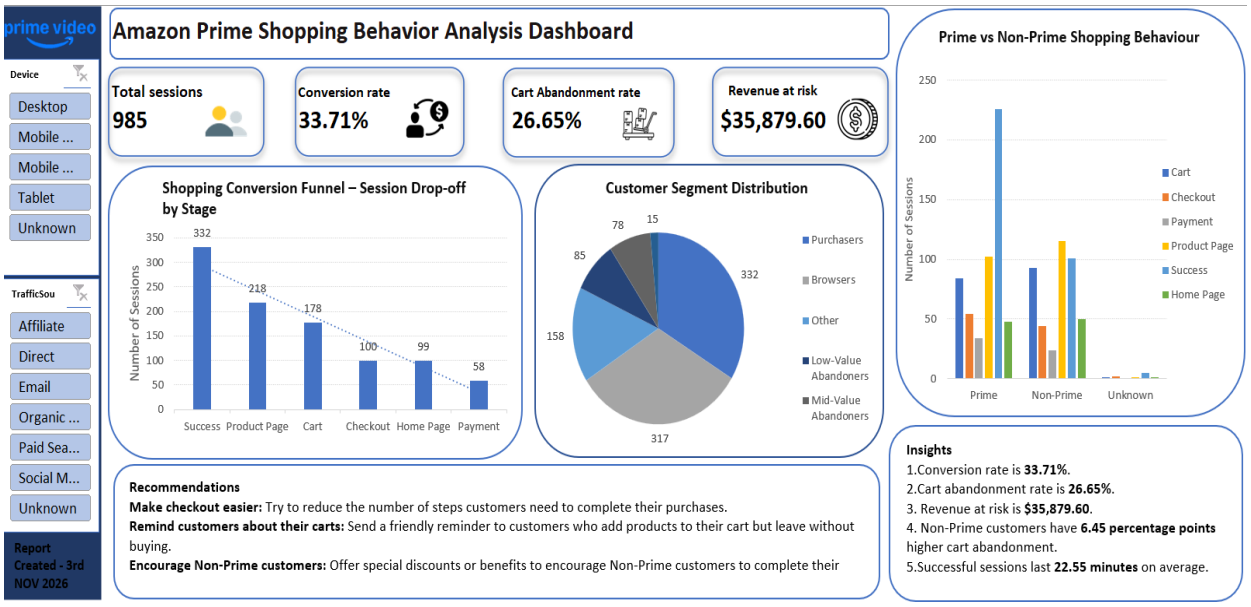


Figure 10: Interactive Shopping Behaviour Analysis Dashboard.

9. Key Findings and Insights

The analysis of shopping behaviour revealed several important findings:

1. **Low Purchase Conversion:** Only 33.71% of the total shopping sessions resulted in successful purchases.
2. **Cart Abandonment:** The cart abandonment rate was 26.65%, showing that a considerable number of customers left after adding products to their carts.
3. **Revenue at Risk:** The value of abandoned carts was $35,879.60, representing a potential opportunity to recover revenue.
4. **Checkout and Payment Drop-offs:** The combined abandonment rate at the Checkout and Payment stages was 32.24%, highlighting an area that may need improvement.
5. **Customer Behaviour:** Successful sessions had an average duration of 22.55 minutes, compared with 9.89 minutes for abandoned sessions.
6. **Membership Behaviour:** Non-Prime customers showed a 6.45 percentage-point higher cart abandonment rate than Prime customers.

10. Recommendations

Based on the findings of the analysis, the following improvements are suggested:

1. **Simplify the Checkout Process**: Reduce unnecessary steps during checkout and make the process easier so customers can complete their purchases without difficulty.
2. **Send Cart Abandonment Reminders:** Send friendly reminders to customers who leave products in their carts. This may encourage them to return and complete their purchases.
3. **Offer Benefits to Non-Prime Customers:** Consider providing special discounts, offers or other benefits to Non-Prime customers, as their cart abandonment rate was higher than that of Prime customers.
4. **Provide Cash on Delivery (COD):** Offer Cash on Delivery as a payment option for both small and high-value purchases, subject to suitable order limits and verification. This may build customer trust, make customers more comfortable placing orders and encourage them to complete their purchases.

These recommendations may help improve the shopping experience on amazon, reduce cart abandonment and increase completed purchases.

11. Conclusion

This project provided an opportunity to understand customer shopping behaviour through data analysis using Microsoft Excel and PostgreSQL.

The dataset was cleaned, organised and analysed to identify cart abandonment patterns, customer segments and potential revenue at risk. Important KPIs were calculated, and the results were verified using both Excel and SQL. An interactive Excel dashboard was also created to present the findings clearly.

The analysis highlighted areas where customers tend to leave the shopping process and showed differences in the behaviour of Prime and Non-Prime customers. Based on these findings, practical recommendations were suggested to improve the checkout experience and encourage customers to complete their purchases.

Overall, this project helped develop practical skills in data cleaning, Excel, SQL querying, data visualisation and interpreting business data.